

Jihyun Jeon

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EDUCATION

Carnegie Mellon University **May 2028**

Master of Science Robotic Systems Development

California Institute of Technology (GPA 3.7/4.0) **Dec 2025**

Computer Science Bachelor of Science

Relevant Coursework : Robot Learning, Deep Learning, Autonomy, Mobility, Systems Engineering, Graphics

SKILLS

Languages: Python, C++, C#, Java, C

Tools & Platforms: ROS2, PyTorch, OpenCV, Scikit-Learn, Git, Docker, Linux, OpenGL, CAD (Solidworks, Fusion360)

PROJECTS

MRSD Capstone, Carnegie Mellon University (*Sponsored by Smith & Nephews*) **September 2026 - Present**

- Engineering a robotic surgical assistant for voice-commanded instrument retrieval and handover
- Defining system requirements for speech recognition, perception, manipulation, and human-safe interaction
- Investigating hardware and autonomy specifications for reliable grasping and ergonomic handoff

EVE, Caltech (*Mobile robot*) **Spring 2024**

- Developed custom wall-following navigation strategy ensuring coverage of all token locations in competition
- Integrated LIDAR, IMU and camera for onboard mapping, localizing and path planning

Tanny Grams, Caltech (*Manipulation*) **Winter 2024**

- Built 5 DOF arm for autonomously solving tangram puzzles with ROS2 and OpenCV
- Achieved ~1cm positioning accuracy and recovery from various human interruptions and execution errors

Snow Shovel Coverage Planning, Caltech **Winter 2024**

- Modified wavefront-based coverage planner to generate feasible snow-shoveling paths under task constraints

RESEARCH

Bot Intelligence Group, CMU Robotics Institute (*Adv: Prof. Jean Oh*) **June 2026 – Present**

- Developing a shared-control robotic teleoperation system to assist people with Parkinson's disease in painting
- Investigating VLA models for intent-aware assistance to predict user intent and preserve user agency
- Planning a formative user study and dataset collection pipeline to identify user needs, evaluate sense of agency, and inform shared-control strategies

Caltech Rover Autonomy Club (*Adv: Dr. Soonjo Chung*) **September 2022 – December 2025**

- Designed and fabricated 6 DOF manipulator capable of carrying 5kg load and performing various operations
- Implemented inverse kinematics-based controls with the ROS Control framework on Raspberry Pi 5
- Led integration of 5 subsystems to build a rover capable of wireless rough terrain traversal and manipulation
- Validated system through field testing and competing in two Canadian International Rover Challenges

Center for Autonomous Systems and Technologies, Caltech (*Adv: Dr. Soonjo Chung*) **June 2024 – November 2024**

- Developed VR interface with Unity for Brain Computer Interface to use eye gaze to gauge user intent
- Explored methods for native Oculus app development, such as ExoPlayer, ROS Java, and GStreamer
- Hosted Bi-weekly update meetings for lab of 20+ people, presented results at the SFP seminar

WORK EXPERIENCE

Intern, Tapblaze **July 2023 – September 2023**

- Designed and programmed a web demo of Good Pizza, Great Pizza using C#, Unity and FTP server connection
- Drove the product development for a mobile game and led weekly game design meetings of 15+ people
- Created 10+ prototypes for a mobile game with a new approach to brick breaker-mechanics with Unity

Intern, Aniai **July 2022 – September 2022**

- Analyzed UX flow and designed UI of a cooking robot dashboard with 10+ pages
- Oversaw creation and packaging of GUI widget library for customizable widgets with PyQt
- Deployed GUI for client and server pages, including operation, testing, data analysis, and calibration